SYSTEMATIC PORTFOLIOS / INDEPENDENT RESEARCH

What does rebalancing earn?

Diversification, concentration and the order of returns

QuantLab Research

20 October 2026

EXACT WEALTH IDENTITY
$D - T$
Diversification minus endpoint concentration

SIMULATION GRID
63
Drift-gap and correlation combinations

MONTE CARLO PATHS
2,000
Per cell; five years; fixed random seed

A positive diversification contribution does not guarantee outperformance of buy and hold. Rebalancing captures one growth component while limiting the accumulation of relative winners. The balance between those two effects determines its frictionless terminal advantage.

Executive findings

This paper separates a mathematical mechanism from a tradable backtest. The exact algebra concerns fractional, costless daily rebalancing at the close, starting at 6 January 2020 close. The earlier research used next-open execution, integer adjusted units and costs, with an earlier cash baseline. Their headline numbers are not interchangeable.

For the 50/50 AAPL/TSLA close-to-close history, the annualized diversification contribution is 4.143 log units times 100, while the endpoint concentration term is 3.697. Their difference is only 0.446 annual log units times 100. Constant mix grows at 45.93% CAGR versus 45.28% for buy and hold. At 75% AAPL, the diversification term stays positive but constant mix loses to buy and hold.

Paired-day permutations preserve both assets' terminal returns and every daily constant-mix gross return factor, so daily constant mix's final wealth is exactly unchanged. Monthly rebalancing is sensitive to the ordering. In 1,000 reordered histories, its CAGR has a 5th–95th percentile range of 44.35%–47.69%, while the historical monthly ordering yields 45.07%.

A five-year geometric-Brownian-motion laboratory varies arithmetic drift gaps and correlation. Greater relative variation increases the diversification term, but a persistent drift advantage can make the concentration term larger. The simulation reports uncertainty and is not calibrated as a forecast for these stocks.

Research status. Retrospective analysis, not an investment recommendation or a verified trading record. Independent institutional-style presentation; no affiliation with an investment firm. Definitions, execution assumptions and limitations are documented inside.

01 / MECHANISM IN PLAIN LANGUAGE

1 Two effects pull in opposite directions

Rebalancing earns relative variation but repeatedly sells exposure to relative winners.

Consider two assets and two ways to invest the same initial capital. A constant-mix policy restores the original weights after every period. Buy and hold buys once and then allows the weights to change with prices. Both stay fully invested, but their exposures evolve differently.

When asset returns differ within a period, the log return of the weighted portfolio exceeds the weighted average of the two asset log returns. Accumulating this difference gives a nonnegative *diversification contribution*. The term is a comparison with a weighted log-return benchmark; it is not, by itself, a comparison with buy and hold.

Buy and hold has a second effect. If one asset substantially outgrows the other over the full horizon, its weight increases. This gives buy and hold a nonnegative *endpoint concentration contribution* relative to that same weighted log-return benchmark. A constant-mix policy does not retain that growing concentration.

Subtracting the second term from the first produces the exact terminal log-wealth difference. Either term can dominate. Trading costs can further reduce a rebalanced policy's result, but they are not needed for it to lose to buy and hold.

Experiment layer	Purpose	Execution convention
Historical identity	Explain the two components	Fractional, no costs; daily close to close
Paired-day permutation	Isolate ordering sensitivity	Same return observations; new ordering
Deterministic controls	Give checkable path examples	Synthetic log shocks; daily / 21-session resets
GBM laboratory	Vary assumptions deliberately	Prescribed drifts, volatilities, correlation
Opening bridge	Retain practical context	Original integer open policy; free / base costs

Evidence boundary. The identity is algebra, the permutation experiment is a conditional rearrangement of one historical sample, and the GBM grid is a hypothetical model. None alone estimates a future investable "rebalancing premium."

The historical mechanism sample starts at the 6 January 2020 close and ends at 31 December 2025 close, giving 1,505 daily return intervals. It deliberately excludes that first session's opening-to-closing return. No integer rounding, fees, tax or auction sizing enters the identity.

02 / EXACT DECOMPOSITION

2 One benchmark makes the comparison precise

The formula separates a period-by-period term from a terminal relative-growth term.

Let $g_{it} > 0$ be asset i's gross return in period t, $w \in (0,1)$ the fixed weight of asset 1, and $L_i = \sum_t \log g_{it}$. Normalize initial wealth to one. Then

$$V_{\mathrm{CM}} = \prod_t \{w g_{1t} + (1-w) g_{2t}\}, \qquad V_{\mathrm{BH}} = w e^{L_1} + (1-w) e^{L_2}.$$

Introduce the weighted log-return benchmark $A = wL_1 + (1-w)L_2$. Define

$$D = \sum_t \left[\log\{w g_{1t} + (1-w) g_{2t}\} - w \log g_{1t} - (1-w) \log g_{2t}\right],$$
$$T = \log\{w e^{L_1} + (1-w) e^{L_2}\} - A.$$

Since $\log V_{\mathrm{CM}} = A + D$ and $\log V_{\mathrm{BH}} = A + T$,

$$\log \frac{V_{\mathrm{CM}}}{V_{\mathrm{BH}}} = D - T.$$

Each summand of D is nonnegative by concavity of log; the same inequality applied to the terminal asset wealths gives $T \geq 0$. Positivity of D therefore does not imply positivity of $D - T$.

Relative returns reveal the structure

Let $z_t = \log g_{1t} - \log g_{2t}$ and define

$$f_w(z) = \log\{(1-w) + w e^z\} - wz.$$

Then $D = \sum_t f_w(z_t)$ while $T = f_w(\sum_t z_t)$. Small period-level relative returns give the local expansion

$$f_w(z) = \tfrac{1}{2} w(1-w) z^2 + O(z^3).$$

Thus squared relative variation can create a positive period-by-period contribution, while a large cumulative relative trend increases the terminal concentration contribution. The expansion is an approximation; the preceding wealth identity is exact.

For diffusion returns with volatilities σ_1, σ_2 and correlation ρ, the leading annual diversification term is

$$\gamma^* \approx \tfrac{1}{2} w(1-w)(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2).$$

This approximates D per year, not the net advantage over buy and hold. In particular, it does not eliminate T, finite-horizon uncertainty or transaction costs.

03 / HISTORICAL ACCOUNTING

3 Positive diversification can nearly cancel out

The endpoint concentration term absorbs much of the historical contribution.

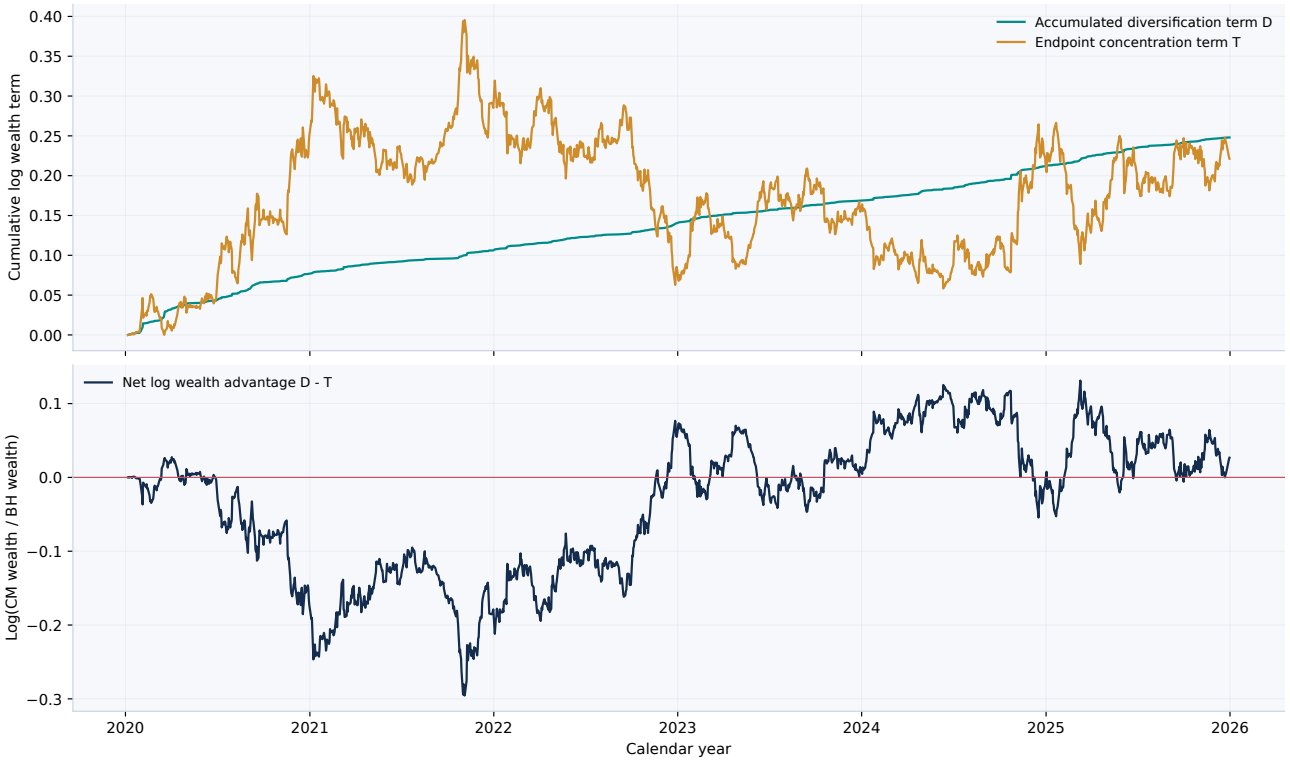


Figure 1: Fractional, frictionless 50/50 close-to-close AAPL/TSLA experiment. Top: accumulated D and endpoint T evaluated through each date. Bottom: their difference, exactly equal to log(constant-mix wealth / buy-and-hold wealth). A negative value means buy and hold has more wealth.

AAPL target	D / year	T / year	Net / year	CM CAGR	BH CAGR
25%	3.105	2.303	0.802	52.99%	51.77%
50%	4.143	3.697	0.446	45.93%	45.28%
75%	3.111	3.526	-0.415	36.34%	36.91%

The three log-contribution columns are annual natural-log wealth units multiplied by 100. They are not arithmetic percentage-point contributions to CAGR. At 50/50, the 0.446 annual log advantage translates into a CAGR difference of about 0.65 percentage points because CAGR depends exponentially on log growth.

At 75% AAPL, D is still positive but T is larger, so constant mix underperforms buy and hold. The identity holds through the entire path with maximum absolute numerical error below 8×10^{-15} across the three weights. This accounting explains the realized terminal comparison without claiming a causal forecast.

04 / WEIGHTS AND REGIMES

4 The same mechanism changes across allocations and years

A term that is nonnegative in level can still decline between dates.

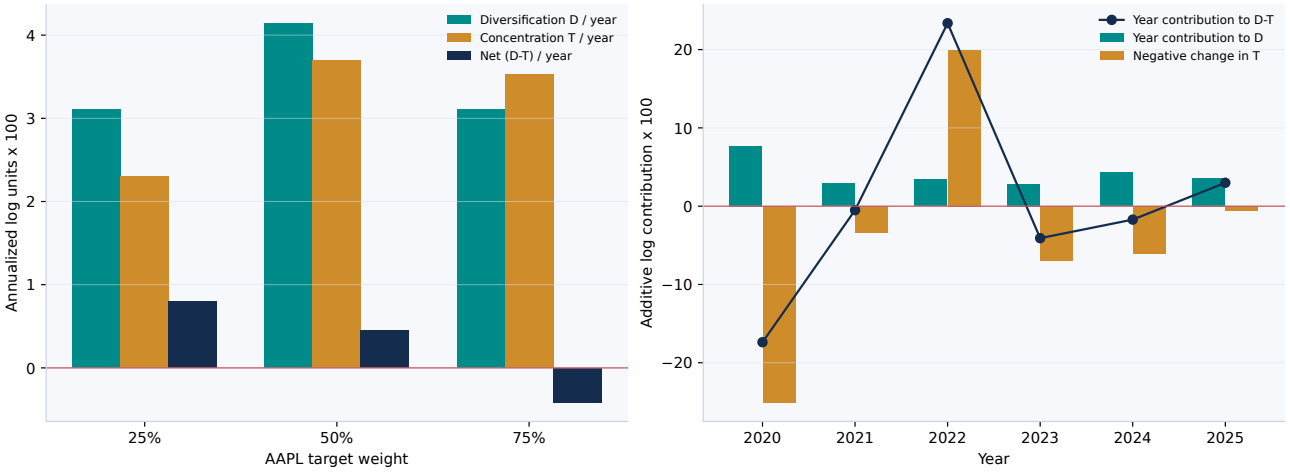


Figure 2: Left: full-history annualized log contributions at three weights. Right: additive calendar-year changes for the 50/50 portfolio, retaining the concentration term accumulated from inception. These are log-wealth contributions, not calendar-year strategy returns.

The local quadratic approximation gives approximately 4.144 annual log units times 100 for the 50/50 diversification contribution, close to the exact 4.143 in this history. That proximity is an observed approximation check, not permission to replace the exact calculation in all markets.

The terminal concentration level T is always nonnegative, but its change over a particular year can be negative. If the cumulative relative winner reverses, the terminal assets become less dispersed and T can fall. Its negative change then contributes positively to the incremental relative performance of constant mix.

The year panel intentionally carries the portfolio from inception. It does not restart both strategies at 50/50 every January. A reset would create a different experiment with a different buy-and-hold reference and different endpoint concentration terms.

The central distinction is between *relative variation within periods* and *cumulative relative growth across the horizon*. Low correlation, high relative volatility or a visibly oscillating price ratio can increase the first quantity, but none fixes the second quantity or guarantees which policy will finish ahead.

Interpretation. "Diversification helps log growth relative to a weighted asset-log benchmark" is a narrower and more precise statement than "rebalancing beats buy and hold." The historical table demonstrates why the distinction matters.

05 / ORDERING EXPERIMENTS

5 Daily terminal wealth is permutation invariant

Monthly rebalancing is sensitive to how the same return observations are grouped.

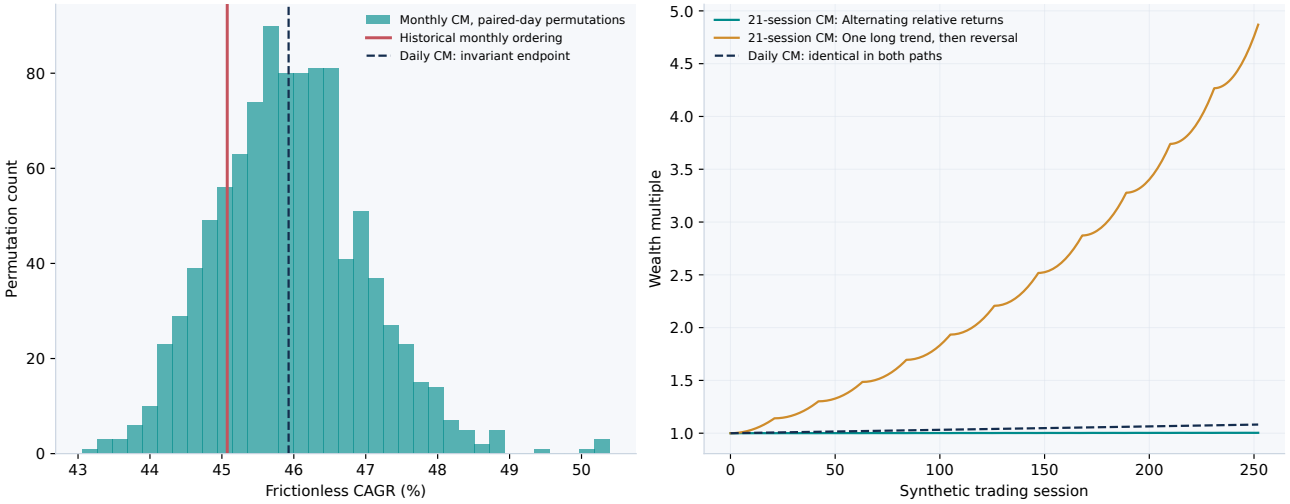


Figure 3: Left: 1,000 paired-day permutations of the historical daily asset returns; monthly dates are fixed. Daily constant-mix final wealth and buy-and-hold final wealth remain unchanged. Right: deterministic synthetic controls with identical asset endpoints and daily constant-mix endpoints, but different 21-session rebalanced wealth.

A permutation moves each pair (g_{1t}, g_{2t}) together to a new date. It preserves both assets' terminal products and the multiset of daily portfolio gross-return factors $wg_{1t} + (1-w)g_{2t}$. Products commute, so daily constant mix and buy and hold have identical terminal wealth across all reorderings. Their drawdown paths need not remain unchanged.

Monthly constant mix instead holds weights between calendar boundaries. Permuting daily observations changes the compounded relative returns inside each holding block. This changes the monthly strategy's final wealth, even though both assets' final wealth and unconditional daily covariance are preserved.

The historical monthly close-based CAGR is 45.07%, at roughly the 19.5th percentile of the rearranged histories. The permutation median is 45.92%, with 5th–95th percentiles of 44.35%–47.69%. These are conditional rearrangement statistics, not a market-generating model or a statistical test that the historical ordering is abnormal.

A transparent synthetic control

For 252 sessions, asset log returns are +0.025 and −0.025 in opposite directions. One sequence alternates signs daily; the other has 126 positive relative sessions followed by 126 negative relative sessions. Both assets end exactly where they began. Daily constant mix ends at 1.0819 times wealth in both cases, while 21-session constant mix ends at 1.0038 and 4.8692, respectively. The shocks are deliberately stylized; they isolate ordering rather than calibrate realistic returns.

06 / SIMULATION LABORATORY

6 Vary drift and correlation deliberately

A model can clarify the mechanism without predicting the next market regime.

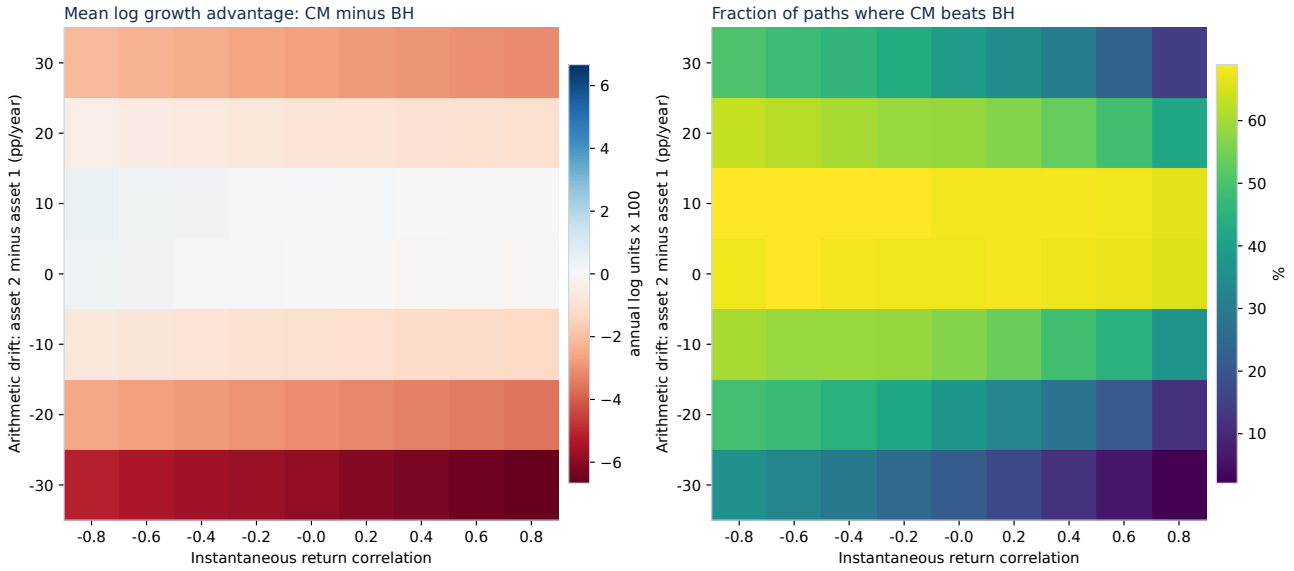


Figure 4: Five-year GBM experiments: 63 cells, 2,000 paths per cell. Mean log advantage and terminal win fraction answer different questions. Each cell uses the same underlying normal draws for controlled comparisons. No trading costs enter.

For each asset, daily log increments are simulated exactly under a GBM specification:

$$\Delta \log S_i = (\mu_i - \tfrac{1}{2}\sigma_i^2)/252 + \sigma_i Z_i/\sqrt{252}.$$

The arithmetic drift center is 8% per year: $\mu_1 = 0.08 - \delta/2$ and $\mu_2 = 0.08 + \delta/2$. The drift gap δ spans −30 to +30 percentage points in ten-point steps. Volatilities are 20% and 40%, correlation ranges from −0.8 to 0.8 in steps of 0.2, and the horizon is 1,260 sessions. The target is 50/50 throughout.

Lower correlation increases relative variation and the average diversification contribution. A large drift gap can nevertheless create a larger concentration term, making buy and hold more attractive in terminal log wealth. Because volatilities differ, equal arithmetic drifts do not imply equal expected log-growth rates or a symmetric net-advantage surface.

Common random numbers are reused across cells, so cell estimates are intentionally dependent. The experiment holds its parameters fixed and does not include changing volatility, jumps, fat tails, momentum states, taxes or execution costs. None of the drift values is presented as an estimate of future AAPL or TSLA returns.

07 / UNCERTAINTY

7 A frequent winner need not have a positive mean advantage

The distribution can be asymmetric, with rare but large relative shortfalls.

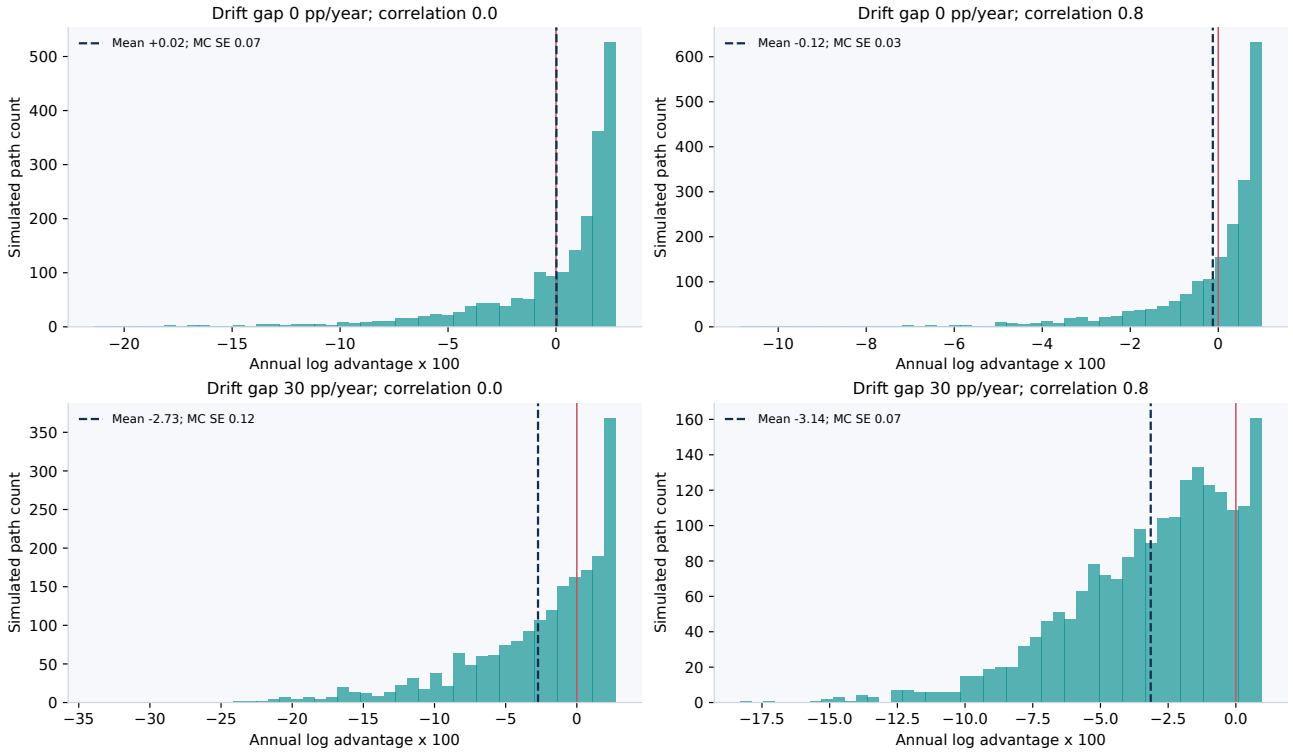


Figure 5: Selected cell distributions of annual log wealth advantage. Red denotes zero; the dashed line is the sample mean. Each legend reports the Monte Carlo standard error of that mean. These distributions are generated by the stated model, not inferred from a large set of independent historical markets.

Drift gap (pp)	Correlation	Mean log x100	MC SE	Win fraction	Win SE (pp)
0	0.0	0.020	0.075	67.5%	1.05
0	0.8	-0.124	0.031	65.2%	1.07
30	0.0	-2.727	0.118	38.9%	1.09
30	0.8	-3.144	0.069	14.6%	0.79

With equal arithmetic drifts and zero correlation, daily constant mix wins in 67.45% of simulated paths, but its mean annual log advantage is only about 0.020 units times 100, with a Monte Carlo SE of 0.075. That mean is numerically close to zero at this simulation precision. The win fraction does not measure the magnitude of gains and losses.

Monte Carlo SE is $s/\sqrt{2000}$ for a cell mean and $\sqrt{p(1-p)/2000}$ for its win fraction. It measures simulation noise conditional on this model. It does not cover parameter uncertainty, model misspecification or future market risk. No formal investment-policy superiority test is claimed.

08 / EXECUTION CONTEXT

8 Bring the mechanism back to the original convention

A frictionless identity and a net trading comparison are different evidence layers.

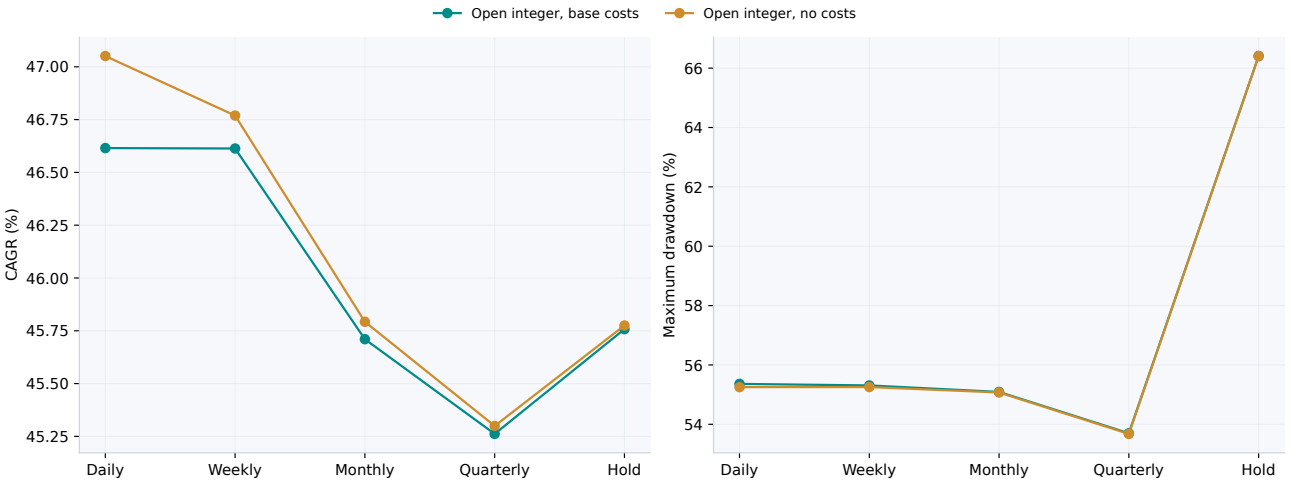


Figure 6: Original historical opening-execution policies with integer adjusted units, rerun with base costs or free execution. Both curves within a panel use the same opening-entry baseline. They are separate from the close-based identity, which starts invested at 6 January close.

Open policy	Free CAGR	Net CAGR	Difference (pp)	Net DD	Base costs
Daily	47.05%	46.62%	0.436	55.36%	$7,857
Weekly	46.77%	46.61%	0.156	55.31%	$2,974
Monthly	45.79%	45.71%	0.083	55.09%	$1,224
Quarterly	45.30%	45.26%	0.038	53.70%	$657
Hold	45.78%	45.76%	0.017	66.41%	$62

The free and base-cost opening runs are paired full simulations. Cost changes quantities and cash as well as direct charges, so their CAGR difference includes the altered compounding path. Slippage attribution is already in fill prices.

The earlier close-based identity is not subtracted from these opening returns to manufacture a timing or cost attribution. Their first invested observation and execution mechanics differ. What transfers is the interpretation: resetting weights changes both diversification exposure and the retention of relative winners, while implementation adds further frictions.

Conclusion. Rebalancing has an exact positive contribution relative to a weighted asset-log benchmark, but its advantage over buy and hold is that contribution minus a second nonnegative term. Costs, holding-period grouping and realized relative trends determine whether the overall policy is attractive. The mechanism supports conditional analysis, not a universal premium claim.

09 / AUDIT

9 Separate proofs, observations and assumptions

All three evidence layers are reproducible, with their limits stated explicitly.

Verification and saved evidence

The exact wealth identity is derived in Section 2 and checked through the full historical path at 25%, 50% and 75% AAPL weights. Maximum residual is below 8×10^{-15}. Each daily diversification summand and each cumulative concentration level is checked for nonnegativity to numerical tolerance. The GBM calculations separately verify the same decomposition path by path.

All 1,000 paired permutations preserve daily constant-mix and buy-and-hold terminal log wealth to 10^{-10}. Deterministic controls have identical endpoint prices and are saved session by session. The simulation seed is 20261020. Complete cell summaries include means, medians, 10th / 90th percentiles and Monte Carlo errors; four selected cells retain all 2,000 endpoint observations. Random paths can be regenerated from the code and recorded NumPy version.

The historical observations come from the unchanged September 2026 Yahoo snapshot used in the original research. Historical identities use adjusted closing prices only; the opening bridge uses the original OHLC adjustment, cost and cash conventions. This is not a second-vendor reconciliation or a point-in-time vintage dataset.

Accounting and metrics

All historical opening-execution runs use USD 100,000 initial cash, two assets, no leverage or shorting, integer adjusted research units and nonnegative cash. Raw OHLC is scaled by adjusted close / raw close. Dividends and splits are not posted again. Adjusted units preserve the original QuantLab convention but are not historical legal share counts; per-unit commissions inherit this limitation.

Base costs are 5 bps adverse opening slippage and USD max(1, 0.005 times quantity) per executed instrument order. Sales precede buys; symbol order breaks purchase ties. Final holdings are marked without liquidation. Slippage is reflected in fills and is not deducted twice. No taxes, cash interest, capacity constraints or intraday drawdown estimates are included.

CAGR uses elapsed calendar days / 365.25. Annual volatility is the sample daily-return standard deviation times $\sqrt{252}$; Sharpe is the arithmetic mean divided by that standard deviation times $\sqrt{252}$, with zero risk-free rate. MDD is the positive magnitude of the largest decline from a running closing peak. Gross turnover sums opening-reference traded notional / pre-trade opening equity and divides by elapsed years, including entry; it is not divided by two. This turnover convention differs from the original engine's prior-close / fill-notional convention.

Files and reproduction

`data/` freezes market inputs and provenance. `results/` contains complete numerical tables, audit checks and compressed paths where applicable. `code/` contains calculation, plotting and report-generation scripts; `figures/` keeps vector PDFs and PNG previews. A SHA-256 manifest records delivered-file integrity. Run from the project directory in the documented Python environment:

```
python code/research.py
python code/validate.py
python code/plot_figures.py
python code/build_report.py
mkdir -p /tmp/quantlab-paper-build
pdflatex -output-directory=/tmp/quantlab-paper-build report.tex
pdflatex -output-directory=/tmp/quantlab-paper-build report.tex
```

Copy the compiled PDF from the temporary build directory. No network is required to rerun frozen-input calculations. The report builder combines computed tables with fixed English interpretations; changed assumptions require a review of those interpretations.

Mechanism-specific metric boundary

The close-based identity uses fractional fully invested portfolios, no costs, 1,505 intervals and the 6 January 2020 close as its initial invested state. Its annualized terms are natural-log wealth divided by elapsed years. The GBM laboratory uses five exact model years and no historical calendar; its mean log advantage is not the mean CAGR difference. Permutations preserve unconditional daily observations while destroying chronology and regime persistence.

The diffusion approximation is derived from the local quadratic expansion; the finite-step identity is evaluated exactly. No external theorem, unexplained proxy or simulated history is presented as a verified tradable return. Positive diversification growth, frequent relative wins and investable net outperformance remain distinct concepts throughout.